MISSING VALUE IDENTIFICATION

Veda Technology – Data Analytics Track

1. Objective

The objective of this task is to identify missing values in a dataset, count them by column, calculate the percentage of missing values, and present the results in a clear summary.

2. Dataset Description

The analysis uses the standard Titanic passenger dataset. It contains passenger-related columns such as age, cabin, and embarkation port. The standard version has 891 rows and 12 columns.

3. Tools Used

Python, Pandas, Matplotlib, Google Colab, and Microsoft Excel.

4. Methodology

• Load the CSV dataset into a Pandas DataFrame.

• Inspect the number of rows, columns, and data types.

• Use isnull().sum() to count missing values in each column.

• Calculate missing percentages by dividing missing counts by the total number of rows and multiplying by 100.

• Create a bar chart and export the summary table for Excel.

5. Missing-Value Summary

Column Name	Missing Values	Missing Percentage
Age	177	19.87%
Cabin	687	77.10%
Embarked	2	0.22%

6. Key Findings

• Cabin has the highest number of missing values: 687 (77.10%).

• Age contains 177 missing values (19.87%).

• Embarked contains 2 missing values (0.22%).

• The missing-value pattern differs by column, so a single treatment may not be suitable for every column.

7. Recommendations

Investigate why values are missing before making changes. Depending on the analysis objective, missing values may be imputed, retained, or rows/columns may be excluded when justified. Do not blindly delete missing values.

8. Conclusion

This task demonstrates how Python and Pandas can be used to identify and summarize missing data. The summary table and chart make it easier to understand which columns require attention. Careful handling of missing values improves data quality and supports more reliable analysis.

Note: The summary figures above refer to the standard Titanic dataset. If a different version of the dataset is used, the counts may differ.